Elements of Geometry in Unitary Spaces

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Contents

Preface

The study of unitary spaces through generalizations of geometrical configurations gives insight to understanding of many concepts in linear spaces and linear algebra. This monograph has been prepared for two essential purposes. First, to give a feeling for the meanings of ideas and concepts. Second, to show how one may generalize an idea which may turn into an interesting research problem.

The monograph has six chapters, as seen in the table of contents. We have tried to keep the continuity in the subject matter in the first five chapters. Chapter Six consists of the study of a few geometrical problems and their generalization, solely for the sake of showing how one may generalize some ideas. The exercises have two basic natures; some are for learning ideas, and others are for research problems.

The reader is expected to have had an elementary course in linear algebra and linear spaces.

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1 A Review of Unitary Spaces

A vector study of a Euclidean plane can be generalized to unitary spaces. Throughout this chapter, we shall give geometric interpretations of many propositions, in a Euclidean plane. In what follows, we only consider vector spaces over the field of real or complex numbers. Vectors will be denoted by Greek letters $\alpha, \beta, \gamma, \cdots$, and scalars by Latin letters $a, b, c, \cdots$.

1.1 Inner Product

Let V be a vector space over the field of complex numbers C. We define a function f on $V \times V$ into C such that:

(1) $f(\eta, \xi) = \overline{f(\xi, \eta)}$, where $\overline{f(\xi, \eta)}$ is the complex conjugate of $f(\xi, \eta)$;

(2) $f(\xi, \xi) \geq 0$ and $f(\xi, \xi) = 0$ if and only if $\xi = \vec{0}$;

(3) $f(a_1\xi_1 + a_2\xi_2, \eta) = a_1 f(\xi_1, \eta) + a_2 f(\xi_2, \eta)$, for all ξ_1, ξ_2, η in V and a_1, a_2 in C.

For convenience, we write $(\xi, \eta) = f(\xi, \eta)$. The norm of a vector ξ will be defined by

$$||\xi|| = (\xi, \xi)^1/2$$

In the case where V is over the field of real numbers, the inner product is commutative; i.e., $(\xi, \eta) = (\eta, \xi)$.

Let V be the Euclidean plane. Then the inner product of ξ and η is

$$(\xi, \eta) = ||\xi|| \, ||\eta|| \cos t$$

where t is the angle between ξ and η. The reader may show that this inner product satisfies (1), (2), and (3).

1.2 Unitary Spaces

A vector space V over the field of real or complex numbers is called *unitary* if an inner product is defined for V. Such a space is also called an *inner product space*.

A unitary space over the field of complex numbers will be denoted by E, and over the field of real numbers by $\Re$. When the space is finite-dimensional, we denote it by E_n if it is complex, and by $\Re_n$ if it is real unitary space. Here, n is the dimension of the space.

We shall state some basic theorems without proof.

1.3 Theorem

Let $\xi \in E$, and c be a complex number. Then

$$||c\xi|| = |c| \, ||\xi||$$

We leave the geometric interpretation in $\Re_2$ as an exercise.

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1.4 Theorem (Schwarz inequality)

Let $\xi, \eta \in E$. Then

$$|(\xi, \eta)| \leq ||\xi|| \ ||\eta||$$

1.5 Theorem (Triangle inequality)

Let $\xi, \eta \in E$. Then

$$||\xi + \eta|| \leq ||\xi|| + ||\eta||$$

1.6 Orthogonality

A vector ξ is orthogonal to η if and only if $(\xi, \eta) = 0$. Since $(\eta, \xi) = 0$ implies $(\eta, \xi) = 0$, whenever ξ is orthogonal to η, then η is also orthogonal to ξ. It is clear that $\vec{0}$ is orthogonal to every vector. A set of vectors is said to be *orthogonal* if and only if each vector of the set is orthogonal to every other vector of the set. An orthogonal set is called *orthonormal* if each vector of the set has norm one.

1.7 Theorem

Every finite set of non-zero orthogonal vectors is linearly independent.

1.8 Bessel's Inequality

Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal set in E, and $\xi \in E$. Then

(1) $|(\xi, \alpha_1)|^2 + \cdots + |(\xi, \alpha_n)|^2 \leq ||\xi||^2$, and

(2) $\eta = \xi - [(\xi, \alpha_1)\alpha_1 + \cdots + (\xi, \alpha_n)\alpha_n]$

is orthogonal to α_i, $i = 1, \cdots, n$. This means n is orthogonal to $[\alpha_1, \cdots, \alpha_n]$; i.e., the subspace spanned by $\alpha_1, \cdots, \alpha_n$.

Geometric interpretation

Let us consider $\Re_2$ and the orthonormal set $\{\alpha_1\}$; i.e., the set consisting of one unit vector (Fig. 1).

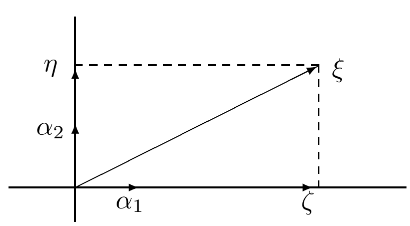

Fig. 1

We observe that $\zeta = (\xi, \alpha_1)\alpha_1$ is the orthogonal projection of ξ on the axis whose unit vector is α_1. Thus

$$||\zeta||^2 = |(\xi, \alpha_1)|^2 \leq ||\xi||^2$$

It is clear that $\eta = \xi - \zeta = \xi - (\xi, \alpha_1)\alpha_1$ is orthogonal to α_1.

If we consider the orthonormal set $\{\alpha_1, \alpha_2\}$, we obtain

$$||\zeta||^2 + ||\eta||^2 = |(\xi, \alpha_1)|^2 + |(\xi, \alpha_2)|^2 = ||\xi||^2$$

1.9 Complete orthonormal set

An orthonormal set of vectors in E is called *complete* in E if and only if the only vector in E that is orthogonal to all elements of the set, is the zero vector.

1.10 Theorem

Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal set in a unitary space E. Then the following statements are equivalent:

(1) $\{\alpha_1, \cdots, \alpha_n\}$ is complete.

(2) If $(\xi, \alpha_i) = 0$ for $i = 1, \cdots, n$, then $\xi = \vec{0}$.

(3) The linear subspace spanned by $\{\alpha_1, \cdots, \alpha_n\}$ is E itself.

(4) For any $\xi \in E$, $\xi = \sum_{i=1}^{n} (\xi, \alpha_i)\alpha_i$.

(5) For a pair $\xi, \eta \in E$, $(\xi, \eta) = \sum_{i=1}^{n} (\xi, \alpha_i)(\alpha_i, \eta)$. (This is called the *Parseval identity*).

(6) For any $\xi \in V$, $||\xi||^2 = \sum_{i=1}^{n} |(\xi, \alpha_i)|^2$.

Geometric interpretation

Let $\{\alpha_1, \alpha_2\}$ be an orthonormal set in $\Re_2$ (Fig. 2).

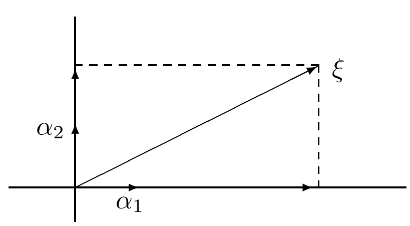

Fig. 2

It is clear that the only vector of the plane orthogonal to α_1 and α_2 is the zero vector. Indeed, $\{\alpha_1, \alpha_2\}$ is a basis for $\Re_2$.

It is clear that ξ is the sum of its two orthogonal projections on the two axes, i.e.

$$\xi = (\xi, \alpha_1)\alpha_1 + (\xi, \alpha_2)\alpha_2$$

Let $\eta \in \Re_2$. Then

$$\eta = (\xi, \alpha_1)\alpha_1 + (\eta, \alpha_2)\alpha_2$$

To explain (5), we observe that the set of components of ξ is

$$((\xi, \alpha_1), (\xi, \alpha_2))$$

The set of components of η is

$$((\eta, \alpha_1), (\eta, \alpha_2))$$

Thus the inner product of ξ and η is

$$\begin{aligned}(\xi, \eta) &= (\xi, \alpha_1)\overline{(\eta, \alpha_1)} + (\xi, \alpha_2)\overline{(\eta, \alpha_2)} \\ &= (\xi, \alpha_1)(\alpha_1, \eta) + (\xi, \alpha_2)(\alpha_2, \eta)\end{aligned}$$

The equality (6) simply indicates that the square of the norm of ξ is the sum of squares of its components (Pythagoras Theorem).

1.11 Theorem

Let E_n be an n-dimensional unitary space, $n \geq 1$. Then there exist complete orthonormal sets in E_n, and every complete orthonormal set contains exactly n elements.

1.12 The Gram-Schmidt process

Let $\{\xi_1, \cdots, \xi_n\}$ be a set of linearly independent vectors in E. Then one can construct a complete orthonormal set $\{\alpha_1, \cdots, \alpha_n\}$ such that each α_i, $i = 1, \cdots, n$ is a linear combination of $\xi_1, \cdots, \xi_n$.

Geometric interpretation

Consider the set of linearly independent vectors $\{\xi_1, \xi_2\}$ in $\Re_2$ (Fig. 3).

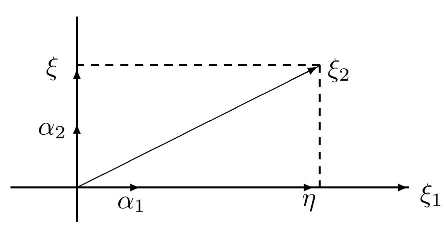

Fig. 3

Here we let α_1 be a unit vector on α_1; i.e., we choose

$$\alpha_1 = \frac{\xi_1}{||\xi_1||}$$

Then we observe that $\xi = \xi_2 - (\xi_2, \alpha_1)\alpha_1$ is orthogonal to α_1. Thus we choose

$$\alpha_2 = \frac{\xi}{||\xi||}$$

In this manner, the set $\{\alpha_1, \alpha_2\}$ is orthonormal, and

$$\alpha_1 = \frac{1}{||\xi_1||}\xi_1$$

$$\alpha_2 = \frac{\xi_2 - (\xi_2, \alpha_1)\alpha_1}{||\xi_2 - (\xi_2, \alpha_1)\alpha_1||}$$

The process

First, for example, we set

$$\alpha_1 = \frac{\xi_1}{||\xi_1||}$$

Then we set

$$\alpha_2 = \frac{\xi_2 - (\xi_2, \alpha_1)\alpha_1}{||\xi_2 - (\xi_2, \alpha_1)\alpha_1||}$$

Now, suppose that $\alpha_1, \cdots, \alpha_k$ have been defined so that they form an orthonormal set and each α_i, $i = 1, \cdots, k$ is a linear combination of $\xi_1, \cdots, \xi_n$. Then we note that

$$\beta_{k+1} = \xi_{k+1} - [(\xi_{k+1}, \alpha_1) + \cdots + (\xi_{k+1}, \alpha_k)]$$

is orthogonal to α_i, $i = 1, \cdots, k$. Thus we set

$$\alpha_{k+1} = \frac{\beta_{k+1}}{||\beta_{k+1}||}$$

Therefore $\{\alpha_1, \cdots, \alpha_{k+1}\}$ is orthonormal, and each element of it is a linear combination of $\xi_1, \cdots, \xi_n$. This establishes the step of the induction, and thus the construction of the set.

Exercise 1

(1) Let α, β be any two distinct elements of an orthonormal set in E. Show that $||\alpha - \beta|| = \sqrt{2}$.

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(2) Let $\{\alpha_1, \alpha_2\}$ be a set of linearly independent vectors in $\Re_2$. Obtain a set of orthogonal vectors $\{\beta_1, \beta_2\}$ in $\Re_2$, such that for each $\xi \in \Re_2$ for which

$$\begin{aligned} \xi &= a_1\alpha_1 + a_2\alpha_2, \\ a_1 + a_2 &= 1, \\ a_1 &> 0, \\ a_2 &> 0 \end{aligned}$$

we also have

$$\begin{aligned} \xi &= b_1\beta_1 + b_2\beta_2, \\ b_1 + b_2 &= 1, \\ b_1 &> 0, \\ b_2 &> 0. \end{aligned}$$

(3) Give a geometric interpretation of (2).

(4) Generalize (2) to n vectors in E_n.

(5) Let $\{\alpha_1, \cdots, \alpha_k\}$ be a set of linearly independent vectors in E. Find a set of orthogonal vectors $\{\gamma_1, \cdots, \gamma_k\}$ which span the same subspace as $\{\alpha_1, \cdots, \alpha_k\}$ does, $||\gamma_i|| = ||\gamma_j||$ for all i, j, and for each

$$\begin{aligned} \xi &= \sum_{i=1}^{k} a_i\alpha_i, \\ \sum_{i=1}^{k} a_i &= 1, \\ a_i &> 0, \\ i &= 1, \cdots, k \end{aligned}$$

we also have

$$\begin{aligned} \xi &= \sum_{i=1}^{k} c_i\gamma_i, \\ \sum_{i=1}^{k} c_i &= 1, \\ c_i &> 0, \\ i &= 1, \cdots, k \end{aligned}$$

(6) Give a geometric interpretation of (5) in $\Re_2$.

(7) Let $\{\alpha_1, \cdots, \alpha_k\}$ be a set of linearly independent vectors in E, and let

$$\xi = \sum_{i=1}^{k} a_i \alpha_i,$$
$$\sum_{i=1}^{k} a_i = 1,$$
$$a_i > 0,$$
$$i = 1, \cdots, k$$

Show that there exists an orthogonal set of vectors $\{\beta_1, \cdots, \beta_k\}$ such that

$$\xi = \sum_{i=1}^{k} b_i \beta_i,$$
$$\sum_{i=1}^{k} b_i = 1,$$
$$b_i > 0,$$
$$i = 1, \cdots, k$$

and

$$\frac{\xi, \beta_i}{||\beta_i||} = \frac{\xi, \beta_j}{||\beta_j||},$$
$$i, j = 1, \cdots, k$$

(8) Give a geometric interpretation of (7) in $\Re_2$.

(9) Show that two vectors $\xi, \eta \in \Re$ are orthogonal if and only if $||\xi+\eta||^2 = ||\xi||^2 + ||\eta||^2$.

(10) Is (9) true if $\xi, \eta \in E$?

(11) Let $||\xi|| = ||\eta|| \neq 0$, and $\xi, \eta \in \Re$. Show that $\xi - \eta$ and $\xi + \eta$ are orthogonal.

(12) Give geometric interpretations for (9) and (11).

(13) Let $\xi, \eta \in E$. Show that

$$||\xi + \eta||^2 + ||\xi - \eta||^2 = 2||\xi||^2 + 2||\eta||^2$$

(14) Give a geometric interpretation of (13) in $\Re_2$.

(15) Let V be a vector space over the field of real numbers, where the norm of $\xi \in V$ is defined (V is called a *normed linear space*). Show that a necessary and sufficient condition for V to be an inner product space with $||\xi||^2 = (\xi, \xi)$, is that

$$||\xi + \eta||^2 + ||\xi - \eta||^2 = 2||\xi||^2 + 2||\eta||^2$$

(This proposition usually employs many properties of the real numbers for its proof.) The reader may look into the geometric interpretation of it in $\Re_2$.

2 Rectilinear Configurations

In this chapter, we give vector proofs for some theorems of geometric nature. Then we generalize them to a unitary space. Indeed, some of the ideas have already been studied in Chapter 1. Most of the proofs are left to the reader.

2.1 Law of cosines

Let $\{\alpha, \beta\}$ be a set of linearly independent vectors in $\Re_2$ (Fig. 4).

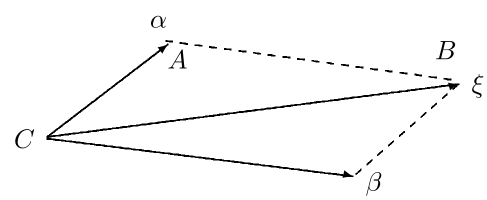

Fig. 4

Let $\xi = \alpha + \beta$. Then we observe that

$$\begin{aligned} ||\xi||^2 &= ||\alpha+\beta||^2 = (\alpha+\beta, \alpha+\beta) \\ &= ||\alpha||^2 + ||\beta||^2 + 2(\alpha, \beta) \end{aligned} \tag{1}$$

If we consider the triangle ABC according to Figure 4, we observe that

$$\begin{aligned} ||\xi|| &= a, \\ ||\alpha|| &= b, \\ ||\beta|| &= c \end{aligned}$$

and

$$(\alpha, \beta) = ||\alpha|| \; ||\beta|| \; \cos t$$

where t is the angle between α and β. Note that the angle A is $\pi - t$. Thus (1) can be written as

$$a^2 = b^2 + c^2 - 2bc \cos A$$

2.2 Theorem (Generalization of 2.1)

Let $\{\alpha_1, \cdots, \alpha_k\}$ be a set of linearly independent vectors in E, and

$$\beta = \sum_{i=1}^{k} \alpha_i$$

Then

$$||\beta||^2 = \sum_{i=1}^{k} ||\alpha_i||^2 + 2 \sum_{i \neq 1}^{k} \Re(\alpha_i, \alpha_j) \tag{2}$$

where $\Re(a)$ means the real part of a. Since the proof is straightforward, we leave it as an exercise.

If $\{\alpha_1, \cdots, \alpha_k\}$ is orthogonal, then (2) becomes

$$||\beta||^2 = \sum_{i=1}^{k} ||\alpha_i||^2$$

which is a generalization of Pythagoras theorem.

2.3 Lemma

Let $x_1, \cdots, x_k$ be k non-zero numbers, real or complex; and let $a_1, \cdots, a_k$ be numbers such that

$$\sum_{i=1}^{k} a_i = 1$$

and

$$a_1 x_1 = \cdots = a_k x_k = y \neq 0$$

Then

$$\frac{1}{y} = \sum_{i=1}^{k} \frac{1}{x_i}$$

The proof is very easy. The lemma has been stated since it will be used often.

2.4 Corollary

Let $a_1, \cdots, a_k$ be positive, and let $x_1, \cdots, x_k$ be real numbers, with

$$\sum_{i=1}^{k} a_i = 1$$

Then $0 \neq a_i x_i < y \neq 0$, $i = 1, \cdots, k$ implies

$$\frac{1}{y} < \sum_{i=1}^{k} \frac{1}{x_i}$$

and $0 \neq a_i x_i > y \neq 0$ implies

$$\frac{1}{y} > \sum_{i=1}^{k} \frac{1}{x_i}$$

The proof is quite easy, and will be omitted.

2.5 Theorem (Altitude of a right triangle)

Let $\{\xi_1, \xi_2\}$ be a set of non-zero orthogonal vectors (Fig. 5).

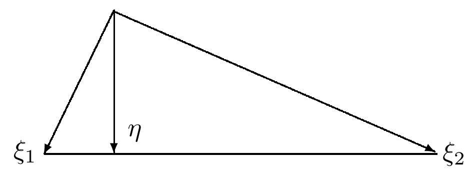

Fig. 5

Let $\eta = a_1\xi_1 + a_2\xi_2$, $a_1 + a_2 = 1$. Then

$$(\eta, \xi_1 - \xi_2) = 0$$

if and only if

$$\frac{1}{||\eta||^2} = \frac{1}{||\xi_1||^2} + \frac{1}{||\xi_2||^2} \tag{3}$$

This means that the vector η is on the altitude of the right triangle formed by ξ_1 and ξ_2 if and only if (3) is true.

Proof

We note that $(\eta, \xi_1 - \xi_2) = 0$ implies that

$$(\eta, \xi_1) = (\eta, \xi_2)$$

Now we observe that

$$(\eta, \xi_1) = a_1||\xi_1||^2 = a_2||\xi_2||^2 = (n, \xi_2) \tag{4}$$

On the other hand,

$$||\eta||^2 = (a_1\xi_1 + a_2\xi_2, a_1\xi_1 + a_2\xi_2) = a_1^2||\xi_1||^2 + a_2^2||\xi_2||^2$$

Employing (4), we obtain

$$\begin{aligned} ||\eta|| &= a_1a_1||\xi_1||^2 + a_2a_2||\xi_2||^2 \\ &= a_1||\xi_1||^2(a_1 + a_2) \\ &= a_1||\xi_1|| \end{aligned}$$

Thus

$$\begin{aligned} \frac{1}{||\xi_1||^2} &= \frac{a_1}{||\eta||^2}, \\ \frac{1}{||\xi_2||^2} &= \frac{a_2}{||\eta||^2} \end{aligned}$$

This implies that

$$\frac{1}{||\xi_1||^2} + \frac{1}{||\xi_2||^2} = \frac{a_1 + a_2}{||\eta||^2} = \frac{1}{||\eta||^2}$$

Now let (3) be true. Suppose $(\eta, \xi_1 - \xi_2) \neq 0$. Then there exists $\zeta = b_1\xi_1 + b_2\xi_2$, $b_1 + b_2 = 1$, such that

$$(\zeta, \xi_1 - \xi_2) = 0$$

Thus by the first part, we have

$$\frac{1}{||\zeta||^2} = \frac{1}{||\xi_1||^2} + \frac{1}{||\xi_2||^2}$$

This implies that

$$||\zeta|| = ||\eta||$$

By (4), we have

$$||\zeta||^2 = b_1||\xi_1||^2 = b_2||\xi_2||^2$$

Thus

$$\begin{aligned}
(\zeta, \eta) &= (b_1\xi_1 + b_2\xi_2, a_1\xi_1 + a_2\xi_2) \\
&= b_1a_1||\xi||^2 + b_2a_2||\xi_2||^2 \\
&= (a_1 + a_2)[b_1||\xi||^2] \\
&= b_1||\xi||^2 = b_2||\xi||^2 = ||\xi||^2
\end{aligned}$$

Also, one obtains

$$(\eta, \zeta) = ||\eta||^2$$

Thus

$$||\zeta - \eta||^2 = (\zeta - \eta, \zeta - \eta) = ||\zeta||^2 - (\zeta, \eta) - (\eta, \zeta) + ||\eta||^2 = 0$$

This implies that $\zeta = \eta$.

2.6 Theorem (Generalization of 2.5)

Let $\{\alpha_1, \cdots, \alpha_k\}$ be an orthogonal set of non-zero vectors in E. Let $\alpha_1, \cdots, \alpha_k$ be positive numbers such that

$$\sum_{i=1}^{k} a_i = 1$$

and

$$\xi = \sum_{i=1}^{n} a_i\alpha_i$$

Then

$$(\xi, \alpha_i - \alpha_j) = 0 \qquad i, j = 1, \cdots, k$$

if and only if

$$\frac{1}{||\xi||^2} = \sum_{i=1}^{k} \frac{1}{||\alpha_i||^2}$$

2.7 Theorem (Angle bisector)

Let $\{\alpha_1, \alpha_2\}$ be a set of non-zero orthogonal vectors in $\Re_2$ (Fig. 6).

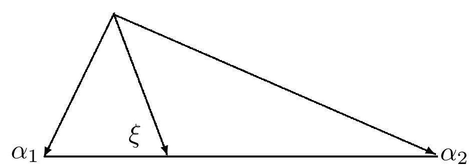

Fig. 6

Let ξ be the vector which represents the bisector of the right angle, i.e.,

$$\xi = a_1\alpha_1 + a_2\alpha_2$$
$$a_1 + a_2 = 1$$

and

$$\left(\xi, \frac{\alpha_1}{||\alpha_1||}\right) = \left(\xi, \frac{\alpha_2}{||\alpha_2||}\right)$$

Then

$$\frac{\sqrt{2}}{||\xi||} = \frac{1}{||\alpha_1||} + \frac{1}{||\alpha_2||}$$

Proof

It is clear that

$$\left(\xi, \frac{\alpha_1}{||\alpha_1||}\right) = \left(\xi, \frac{\alpha_2}{||\alpha_2||}\right)$$

implies

$$a_1||\alpha_1|| = a_2||\alpha_2|| \tag{5}$$

Now, we observe that

$$||\xi||^2 = (a_1\alpha_1 + a_2\alpha_2,\ a_1\alpha_1 + a_2\alpha_2) = a_1^2||\alpha_1||^2 = a_2^2||\alpha_2||^2$$

Considering (5), we obtain

$$||\xi||^2 = 2a_1^2||\alpha_1||^2 = 2a_2^2||\alpha_2||^2$$

Thus

$$||\xi|| = \sqrt{2}a_1||\alpha_1|| = \sqrt{2}a_2||\alpha_2||$$

which implies

$$\frac{\sqrt{2}}{||\xi||} = \frac{1}{||\alpha_1||} + \frac{1}{||\alpha_2||}$$

2.8 Theorem (Equisector)

Let $\{\alpha_1, \cdots, \alpha_k\}$ be a set of non-zero orthogonal vectors in E. Let $a_1, \cdots, a_k$ be positive numbers such that

$$\sum_{i=1}^{k} a_1 = 1$$

and

$$\xi = \sum_{i=1}^{k} a_i\alpha_i$$

Then

$$(\xi, \frac{\alpha_i}{||\alpha_i||}) = (\xi, \frac{\alpha_j}{||\alpha_j||}),\ i, j = 1, \cdots, k$$

implies

$$\frac{\sqrt{k}}{||\xi||} = \sum_{i=1}^{k} \frac{1}{||\alpha_1||}$$

The proof follows the one of 2.7.

2.9 Theorem

Let $\{\alpha_1, \alpha_2\}$ be a set of linearly independent vectors in $\Re_2$. Let the angle between α_1 and α_2 be $\frac{2\pi}{3}$ (Fig. 7).

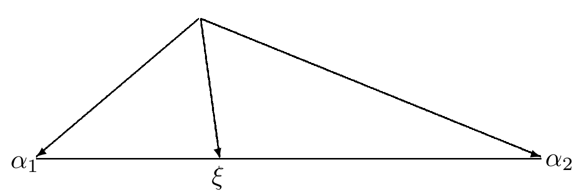

Fig. 7

Let ξ be on the bisector of the angle between α_1 and α_2, and end on the line through the endpoints of α_1 and α_2. Then

$$\frac{1}{||\xi||} = \frac{1}{||\alpha_1||} + \frac{1}{||\alpha_2||}$$

The proof is quite simple.

2.10 Corollary

Let α_1, α_2, and ξ be the same as in 2.9. Let β_1 and β_2 be orthogonal projections of α_1 and α_2 on the line containing ξ (Fig. 8). Then

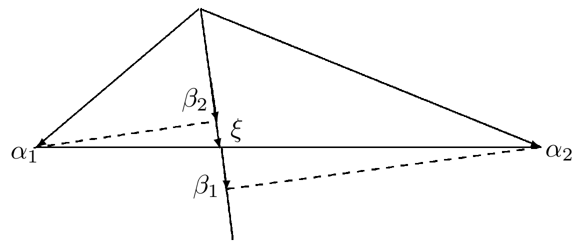

Fig. 8

$$\frac{2}{||\xi||} = \frac{1}{||\beta_1||} + \frac{1}{||\beta_2||}$$

Here, the end point of ξ is called the *harmonic conjugate* of the origin with respect to end points of β_1 and β_2. The proof is left as an exercise.

2.11 Theorem (Generalization of 2.9)

Let $\{\alpha_1, \cdots, \alpha_k\}$ be a set of linearly independent vectors in $\Re$. Let $a_1, \cdots, a_k$ be positive numbers such that

$$\sum_{i=1} k a_i = 1$$

and

$$\xi = \sum_{i=1}^{k} a_i \alpha_i$$

Let

$$\left(\frac{\alpha_i}{||\alpha_i||}, \frac{\alpha_j}{||\alpha_j||} \right) = -\frac{1}{k}, \quad i, j = 1, \cdots, k, \quad i \neq j$$

and

$$\left(\xi, \frac{\alpha_i}{||\alpha_i||} \right) = \left(\xi, \frac{\alpha_j}{||\alpha_j||} \right), \quad i, j = 1, \cdots, k$$

Then

$$\frac{1}{||\xi||} = \sum_{i=1}^{k} \frac{1}{||\alpha_1||}$$

2.12 Corollary

Let α_i and ξ be the same as in 2.11.

Define

$$\beta_i = \left(\alpha_i,\ \frac{\xi}{||\xi||}\right)\frac{\xi}{||\xi||}$$

Then

$$\frac{k}{||\xi||} = \sum_{i=1}^{k}\frac{1}{||\beta_i||}$$

The vector ξ is defined to be the *harmonic conjugate* of the zero vector with respect to $\{\beta_1, \cdots, \beta_k\}$.

Exercise 2

(1) Let $\{\alpha_1, \alpha_2\}$ be linearly independent in $\Re_2$, and

$$\xi = a_1\alpha_1 + a_2\alpha_2,$$
$$a_1 + a_2 = 1$$

(i) Show that there is a unique vector $\xi \in \Re_2$ such that $(\xi, \alpha_1 - \alpha_2) = 0$.

(ii) What is the geometric meaning of (i)?

(iii) Obtain an expression for $||\xi||$ independent of a_1 and a_2.

(iv) Show that there exists $\eta \in \Re_2$ such that

$$\left(\eta, \frac{\alpha_1}{||\alpha_1||}\right) = \left(\eta, \frac{\alpha_2}{||\alpha_2||}\right)$$

(v) Give a geometric interpretation of (iv).

(vi) Obtain an expression for $||\eta||$ independent of a_1 and a_2.

(vii) Show that there is $\zeta \in \Re_2$ such that

$$||\zeta - \alpha_1|| = ||\zeta - \alpha_2||$$

(viii) Give a geometric interpretation of (vii).

(ix) Find $||\zeta||$ in terms $||\alpha_1||$, $||\alpha_2||$, and $||\xi||$.

(2) Generalize (1) to $\Re_n$.

(3) Generalize (1) to E_n.

(4) Generalize 2.11 and 2.12 to the complex spaces.

(5) Let $\{\alpha_1, \alpha_2\}$ be a set of non-zero orthogonal vectors in $\Re_2$.
Let $\xi = a_1\alpha_1 + a_2\alpha_2$ with $a_1 + a_2 = 1$ and $(\xi, \alpha_1 - \alpha_2) = 0$.
Show that

$$||\alpha_1||\ ||\alpha_2|| = ||\xi||\ ||\alpha_1 + \alpha_2||$$

(6) Give a geometric interpretation of (5).

(7) In (5), show that $||\alpha_1+\alpha_2||$ is the diameter of the circumscribed circle of the triangle formed by α_1 and α_2.

(8) Generalize (5) to the case where $\{\alpha_1, \alpha_2\}$ is linearly independent.

(9) Generalize (5) to $\Re_n$, and obtain inequalities.

(10) Study other properties of a right triangle, and generalize the idea to a set of non-zero orthogonal vectors in E.

3 The Space of Linear Transformations

In this chapter, we study some metric properties of linear transformations on a unitary space E_n of dimension n.

3.1 Preliminaries

Let A be a linear transformation on E_n. Then the adjoint of A is denoted by A^*, and is defined by

$$(A\xi, \eta) = (\xi, A^*\eta),$$
$$\xi, \eta \in E_n$$

If $A^* = A$, then A is called a *Hermitian transformation* or a *self-adjoint transformation*. If $A^*A = AA^*$, then A is called a *normal transformation*. If $A^*A = AA^* = I$, then A is a *unitary transformation*. A Hermitian transformation A is called *positive* (*positive definite*) if for every $\xi \neq 0$, $\xi \in E_n$, we have $(A\xi, \xi) > 0$. A Hermitian transformation A is called *non-negative* (*positive semidefinite*) if for every $\xi \in E$, we have $(A\xi, \xi) \geq 0$.

Now, we shall review a few theorems without proof. Proper values of a positive (non-negative) transformation are positive (non-negative). Let $m_1, \cdots, m_n$ be proper values of a Hermitian transformation A. Then there exists an orthonormal basis $\{a_1, \cdots, a_n\}$ such that

$$A\alpha_i = m_i a_i,$$
$$i = 1, \cdots, n$$

Whenever necessary, we shall state other theorems.

3.2 Theorem (Fischer)

Let A be a Hermitian transformation on E_n with proper values $m_1 > \cdots > m_n$. Then

$$m_1 = \sup_{||\xi_1||=1} (A\xi, \xi) \tag{1}$$

and

$$m_n = \inf_{||\xi_1||=1} (A\xi, \xi) \tag{2}$$

Proof

Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal set of proper vectors of A such that

$$A\alpha_i = m_i\alpha_i,$$
$$i = 1, \cdots, n$$

Then for any $\xi \in E_n$ by 1.10 (4), we have

$$\xi = (\xi, \alpha_1)\alpha_1 + \cdots + (\xi, \alpha_n)\alpha_n$$

and

$$A\xi = m_1(\xi, \alpha_1)\alpha_1 + \cdots + m_n(\xi, \alpha_n)\alpha_n$$

Therefore

$$(A\xi, \xi) = m_1|(\xi, \alpha_1)|^2 + \cdots + m_n|(\xi, \alpha_n)|^2$$

To prove (1), we observe that

$$(A\xi, \xi) \leq m_1[|(\xi, \alpha_1)|^2 + \cdots + |(\xi, \alpha_n)|^2]$$

Now if we let $||\xi|| = 1$, by 1.10 (6), we have

$$1 = ||\xi||^2 = |(\xi, \alpha_1)|^2 + \cdots + |(\xi, \alpha)|^2$$

Therefore

$$(A\xi, \xi) \leq m_1$$

But when $\xi = \alpha_1$, we obtain

$$(A\xi, \xi) = (A\alpha_1, \alpha_1) = (m_1\alpha_1, \alpha_1) = m_1$$

Thus

$$\sup_{||\xi||=1} (A\xi, \xi) = m_1$$

Now to prove (2), one notes that

$$(A\xi, \xi) \geq m_n[|(\xi, \alpha_1)|^2 + \cdots + |(\xi, \alpha_n)|^2]$$

If we let $||\xi|| = 1$, we obtain

$$(A\xi, \xi) > m_n$$

Again for $\xi = \alpha_n$, we have

$$(A\xi, \xi) = (A\alpha_n, \alpha_n) = (m_n\alpha_n, \alpha_n) = m_n$$

Consequently

$$\inf_{||\xi||=1} (A\xi, \xi) = m_n$$

Many generalizations may be found in the literature. Since we do not use them, we shall not pursue that direction.

3.3 Theorem

Let A be a linear transformation on E_n. Then A^*A and AA^* are both non-negative, and have the same proper values.

3.4 Definition (Square roots)

We shall only define the non-negative square roots for non-negative transformations. Let A be a non-negative transformation on E_n with proper values $h_1 \geq \cdots \geq h_n$. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis in E_n such that

$$A\alpha_i = h_i\alpha_i,$$
$$i = 1, \cdots, n$$

Then the matrix of A with respect to this basis is

$$\begin{pmatrix} h_1 & & 0 \\ & \ddots & \\ 0 & & h_n \end{pmatrix}$$

which is in diagonal form. We define the linear transformation B on E_n whose matrix with respect to $\{\alpha_1, \cdots, \alpha_n\}$ is

$$\begin{pmatrix} \sqrt{h_1} & & 0 \\ & \ddots & \\ 0 & & \sqrt{h_n} \end{pmatrix}$$

Then B is called the *non-negative square root* of A, and will be denoted by $\sqrt{A}$. It is clear that $\sqrt{A}$ is non-negative, and has the same proper vectors as A.

3.5 Definition (Singular values)

Let A be a linear transformation on E_n, and let $a_1, \cdots, a_n$ be proper values of A^*A. Then the set $\{\sqrt{a_1}, \cdots, \sqrt{a_n}\}$ is called the set of *absolute singular values* of A. Note that we have only considered positive square roots. In what follows, whenever we say singular values, we mean absolute singular values.

Here, we may easily show that the absolute singular values of A are the same as the proper values of $\sqrt{A^*A}$ or the proper values of $\sqrt{AA^*}$. We leave the proof as an exercise.

3.6 The Hilbert Norm

Let A be a linear transformation on E_n. Then the Hilbert norm of A is defined by

$$||A||_H = \sup_{||\xi||=1} ||A\xi||$$

One can express this norm in terms of singular values of A. Let a be the largest singular value of A. Then we observe that a^2 is a proper value of A^*A. Then by (3.2) (1), we have

$$\sup_{||\xi||=1} (A^*A\xi, \xi) = a^2$$

But

$$(A^*A\xi, \xi) = (A\xi, A\xi) = ||A\xi||^2$$

Therefore

$$||A||_H^2 = \sup_{||\xi||=1} ||A\xi||^2 = a^2$$

This implies that

$$||A||_H = a$$

3.7 Frobenius inner product

Let A and B be linear transformations on E_n. Let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis in E_n. Let (a_{ij}) and (b_{ij}) be respectively the matrices of A and B with respect to $\mathcal{B}$. Then we define the Frobenius inner product of A and B by

$$(A, B)_F = \sum_{i,j=1}^{n} a_{ij}\bar{b}_{ij}$$

For example, let

$$[A] = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \cdots & & \cdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}$$

and

$$[B] = \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \cdots & & \cdots \\ b_{n1} & \cdots & b_{nn} \end{pmatrix}$$

Then

$$(A, B)_F = a_{11}\bar{b}_{11} + a_{12}\bar{b}_{12} + \cdots + a_{nn}\bar{b}_{nn}$$

This inner product introduces the Frobenius norm for A; i.e.,

$$||A||_F^2 = (A, A)_F = \sum_{i,j=1}^{n} (a_{ij})^2$$

The next theorem will show that this inner product is well-defined; i.e., it is independent of the choice of the basis.

3.8 Theorem

Let A and B be linear transformations on E_n. Then

$$(A, B)_F = \text{trace } AB^*$$

3.9 Theorem

Let A be a linear transformation on E_n. Let $a_1, \cdots, a_n$ be singular values of A. Then

$$||A||_F = \left(\sum_{i=1}^{n} a_i^2\right)^{1/2}$$

Exercise 3

(1) Obtain all the singular values of

(i)
$$\begin{pmatrix} 1 & i \\ 0 & 3 \end{pmatrix}$$

(ii)
$$\begin{pmatrix} 2i & -2i \\ -1 & 1 \end{pmatrix}$$

(2) Find the singular values of

$$\begin{pmatrix} \frac{1-i}{2} & \frac{1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{pmatrix}$$

(3) Let A be a Hermitian transformation with proper values $m_1 \geq \cdots \geq m_n$. Prove that

$$m_k = \sup_{\substack{M \\ \dim M = k}} \inf_{\substack{||\xi||=1 \\ \xi \in M}} (A\xi, \xi)$$

where M ranges over all subspaces of dimension k of E_n. (This proposition is called *Fischer's Minimax principle.*)

(4) Consider the Frobenius inner product. Express Schwartz inequality in terms of proper values and singular values of A, B, A^*, B^*, or other combinations.

(5) Using Frobenius inner product, obtain the law of cosines.

(6) Generalize 2.5 for the space of linear transformations, using Frobenius inner product.

(7) Generalize (6) for k linear transformations.

(8) Generalize 2.7 for the space of linear transformations, using Frobenius inner product.

(9) Generalize (8) for k linear transformations.

(10) Generalize 2.9 for linear transformations, using Frobenius inner product.

(11) Generalize 2.10 for linear transformations, using Frobenius inner product.

(12) Give generalizations of (10) and (11) for k linear transformations.

4 Some Geometric Properties of Linear Transformations

The Hilbert norm for linear transformations does not induce an inner product on the space of linear transformations. But we may use other kinds of products, and study many configurations of geometric nature consisting of linear transformations.

4.1 Adjoint products

Let A and B be linear transformations. We define $A * B$ to be the adjoint product or $*$-product of A and B. This product does not have all the properties of an inner product. But one may study problems of geometrical natures using this product.

4.2 Theorem

Let $\{A_1, \cdots, A_k\}$, $k < n$, be a set of linear transformations on E_n with a common null space, such that

$$\begin{gathered} A_i * A_j = 0 \\ i < j \\ i, j = 1, \cdots, k \end{gathered}$$

Let $a_1, \cdots, a_k$ be positive numbers such that
Suppose

$$m_1 \geq \cdots \geq m_p > m_{p+1} = \cdots = m_n = 0$$

are singular values of B, and

$$h_{i1} \geq \cdots \geq h_{ip} > h_{i(p+1)} = \cdots = h_{in} = 0 \tag{1}$$

be singular values of A_i, $i = 1, \cdots, k$.

Then

$$\begin{gathered} B * (A_i - A_j) = 0, \\ i, j = 1, \cdots, k \end{gathered} \tag{2}$$

if and only if

$$\begin{gathered} \frac{1}{m_q^2} = \sum_{i=1}^{k} \frac{1}{h_{iq^2}}, \\ q = 1, \cdots, p \end{gathered}$$

(Note that this is another generalization of 2.5.)

4.3 Theorem

Let $\{A_1, \cdots, A_k\}$, $k < n$ be a set of linear transformations on E_n with a common null space such that

$$A_i * A_j = 0, \quad i < j, \quad i,j = 1, \cdots, k$$

Let $a_1, \cdots, a_k$ be positive numbers such that

$$\sum_{i=1}^{k} a_i = 1,$$
$$B = \sum_{i=1}^{k} a_i A_i$$

Suppose

$$m_1 \geq \cdots \geq m_p > m_{p+1} = \cdots = m_n = 0$$

are singular values of B, and

$$h_{i1} \geq \cdots \geq h_{ip} > h_{i(p+1)} = \cdots = h_{in} = 0$$

are singular values of A_i, $i = 1, \cdots, k$.

Then

$$\frac{1}{h_{iq}} B * A_i = \frac{1}{h_{jq}} B * A_j, \quad i,j = 1, \cdots, k \tag{3}$$

and $q = 1, \cdots, p$ implies that

$$\frac{\sqrt{k}}{|m_q|} = \sum_{i=1}^{k} \frac{1}{h_{iq}}$$

(Note that this theorem is another generalization of 2.7.)

4.4 Definition (Reciprocals)

We shall only define reciprocals (generalized inverses) for Hermitian transformations. Let A be a Hermitian transformation on E_n with proper values

$$h_1 \geq h_2 \geq \cdots \geq h_n$$

Let, for example,

$$h_k = h_{k+1} = \cdots = h_p = 0$$

and the rest be non-zero. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for E_n such that

$$A\alpha_i = h_i\alpha_i, \quad i = 1, \cdots, n$$

Thus the matrix of A with respect to this basis is

$$\begin{pmatrix} h_1 & & & & & & & & 0 \\ & \ddots & & & & & & & \\ & & h_{k-1} & & & & & & \\ & & & 0 & & & & & \\ & & & & \ddots & & & & \\ & & & & & 0 & & & \\ & & & & & & h_{p+1} & & \\ & & & & & & & \ddots & \\ 0 & & & & & & & & h_n \end{pmatrix}$$

We define the linear transformation B on E_n whose matrix with respect to $\{\alpha_1, \cdots, \alpha_1\}$ is

$$\begin{pmatrix} 1/h_1 & & & & & & & & 0 \\ & \ddots & & & & & & & \\ & & 1/h_{k-1} & & & & & & \\ & & & 0 & & & & & \\ & & & & \ddots & & & & \\ & & & & & 0 & & & \\ & & & & & & 1/h_{p+1} & & \\ & & & & & & & \ddots & \\ 0 & & & & & & & & 1/h_n \end{pmatrix}$$

It is clear that B is Hermitian, and has the same proper vectors as A. We observe that

$$AB = BA = P$$

where P is the orthogonal projection on the range of A. Here we define B to be the *reciprocal* of A, and write $B = A^-$. It is clear that when A is non-singular, then $A^- = A^{-1}$, the inverse of A.

4.5 Theorem

Here we formulate Theorem 4.3 in terms of transformations. Let $\{A_1, \cdots, A_k\}$, $k < n$ be a set of linear transformations on E_n, with a common null space such that

$$A_i * A_j = 0,$$
$$i < j,$$
$$i, j = i, \cdots, k$$

Let $a_1, \cdots, a_k$ be positive numbers such that

$$\sum_{i=1}^{k} a_i = 1,$$

$$B = \sum_{i=1}^{k} a_i A_i$$

Then

$$B * A_i \left(\sqrt{A_i * A_i}\right)^{-} = B * A_j \left(\sqrt{A_j * A_j}\right)^{-}, \quad i, j = 1, \cdots, k \tag{4}$$

implies that

$$\frac{\sqrt{k}}{m_q} = \sum_{i=1}^{k} \frac{1}{h_{iq}}$$

where m_q and h_{iq}, $i = 1, \cdots, k$, $q = 1, \cdots, p$ are the same as in 4.3.

Exercise 4

(1) Obtain an analogue of 2.9 using $*$-product.

(2) Generalize 2.10 using $*$-product.

(3) Generalize 2.11 for linear transformations using $*$-product.

(4) Using $*$-product, generalize 2.12.

(5) Using $*$-product, what can be said similar to Schwarz inequality?

(6) Same as (5) for Triangle inequality.

(7) Using $*$-product, obtain an analogue of Bessel's inequality.

(8) Obtain an analogue of Parseval identity using $*$-product.

(9) Supply a proof for 4.5.

(10) Let ξ be a vector in E_n such that $||\xi|| = a > 0$. Then ξ describes a sphere in E_n. Using the ideas of norm and $*$-product, define a sphere in the space of linear transformations.

(11) In (10), the sphere had its center at the $\vec{0}$. Write the equation of the sphere whose center is described by $\alpha \epsilon E_n$.

(12) Do (11) for linear transformations.

5 Quadrics

As the reader is already acquainted, the theory of quadratic forms with real variables is closely related to the theory of symmetric matrices and Hermitian transformations. In this chapter, we shall review these ideas. Then we extend the theory to *quadrics*.

5.1 Quadratic forms in two variables

Let x and y be two real variables, and a, b, c be real numbers. Then we observe that the quadratic form

$$q = ax^2 + 2bxy + cy^2$$

can be written as

$$(q) = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Let $\{\alpha_1, \alpha_2\}$ be the orthonormal basis with respect to which q has been written. Then we may say $\xi = x\alpha_1 + y\alpha_2$, and there is a linear transformation A on $\Re_2$ whose matrix is

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

Thus $A\xi$ is represented by

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

And q can be written as

$$(A\xi, \xi) = (\xi, A\xi) \tag{1}$$

Therefore, in general, given a linear transformation A on E_n, we define $(A\xi, \xi)$ to be the quadratic form corresponding to A. The equality (1) can be written if and only if A is Hermitian.

5.2 Reduction of a quadratic form

Let

$$q = (A\xi, \xi) \tag{2}$$

be a quadratic form on E_n, where A is Hermitian. It is well-known that there is an orthonormal basis in E_n with respect to which q has the form

$$q = m_1 y_1^2 + \cdots + m_n y_n^2 \tag{3}$$

where $m_1, \cdots, m_n$ are proper values of A. The form (3) is called the *canonical form* or the *reduced form* of (2).

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5.3 Quadrics in R_n

Let us look at the equation

$$ax^2 + 2bxy + cy^2 + 2rx + 2sy + t = 0 \tag{4}$$

This is the equation of a conic section in its general form. Now we shall consider homogeneous coordinates. A position vector ξ corresponds to a row matrix $\begin{pmatrix} x & y & 1 \end{pmatrix}$, and a direction vector corresponds to $\begin{pmatrix} \ell & m & 0 \end{pmatrix}$. Thus (4) can be written as

$$\begin{pmatrix} x & y & 1 \end{pmatrix} \begin{pmatrix} a & b & r \\ b & c & s \\ r & s & t \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \end{pmatrix} \tag{5}$$

Finally, here we can write (5) in the form of

$$(A\xi, \xi) = 0 \tag{6}$$

where ξ is represented in its homogeneous form.

Note that the inner product should be treated carefully. So in general, (6) is defined to be a quadric, where ξ is represented by

$$(x_1 \cdots x_n 1)$$

and $(x_1, \cdots, x_n)$ is the set of components of ξ with respect to an orthonormal basis in $\Re$.

In what follows, A represents the matrix of the quadric. Each quadric has a quadratic form. For example, in (4),

$$ax^2 + abxy + cy^2$$

is the quadratic form of it. We shall denote the matrix of the quadratic form of a quadric by Q. Let δ be a direction vector. One can easily show that

$$(A\delta, \delta) = (Q\delta, \delta)$$

where in the left-hand side, δ is given by $(\ell_1 \cdots \ell_n)$, and in the right-hand side by the matrix $(\ell_1 \cdots \ell_n)$.

5.4 Straight lines in R_n

It is well-known that the parametric equation of a straight line in $\Re_2$ is

$$\begin{cases} x = x_0 + t\ell y = y_0 + tm \end{cases} \tag{7}$$

where (x_0, y_0) represents a fixed point and (ℓ, m) represents a fixed direction.

It is clear that (7) can be written as

$$\xi = \eta + t\delta$$

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where ξ is the vector whose set of components is (x, y), η corresponds to (x_0, y_0), and δ stands for (ℓ, m).

In general, we may write

$$\xi = \eta + t\delta \tag{8}$$

for a straight line in $\Re_n$. If $\{\alpha_1, \cdots, \alpha_n\}$ is an orthonormal basis, then

$$\begin{aligned} \xi &= x_1\alpha_1 + \cdots + x_n\alpha_n, \\ \eta &= x_{01}\alpha_1 + \cdots + x_{0n}\alpha_n, \\ \delta &= \ell_1\alpha_1 + \cdots + \ell_n\alpha_n \end{aligned}$$

We may write (8) as

$$(x_1 \cdots x_n \ 1) = (x_{01} \cdots x_{0n} \ 1) + t(\ell_1 \cdots \ell_n \ 0)$$

in its homogeneous form. Note that the direction vector δ has been expressed by $(\ell_1 \cdots \ell_n \ 0)$.

5.5 Intersection of a straight line and a quadric

Let $(A\xi, \xi) = 0$ be a quadric, and $\xi = \eta + t\delta$ be a straight line in $\Re_n$. Then the points of intersection of the two are solutions of

$$\begin{cases} (A\xi, \xi) = 0 \\ \xi = \eta + t\delta \end{cases}$$

Eliminating ξ, we obtain

$$(A\delta, \delta)t^2 + 2(A\eta, \delta)t + (A\eta, \eta) = 0$$

This can be written as

$$(Q\delta, \delta)t^2 + 2(A\eta, \delta)t + (A\eta, \eta) = 0 \tag{9}$$

Clearly (9) is a second degree equation in t, and one may consider different cases:

(a) If $(Q\delta, \delta) \neq 0$, then the line intersects the quadric in two points, real or complex, conjugate or coincident.

(b) If $(Q\delta, \delta) = 0$, $(A\eta, \delta) \neq 0$, then the line and quadric intersect in one point.

(c) If $(Q\delta, \delta) = 0$, $(A\eta, \delta) = 0$, and $(A\eta, \eta) \neq 0$, then the line does not intersect the quadric.

(d) If (9) is an identity, then the line is on the quadric.

5.6 A center of a quadric

A vector η is said to end at a center of a quadric $(A\xi, \xi) = 0$ if for any ξ_1 satisfying $(A\xi_1, \xi_1) = 0$, there is a ξ_2 satisfying $(A\xi_2, \xi_2) = 0$ such that

$$\eta = \frac{1}{2}(\xi_1 + \xi_2) \tag{10}$$

The reader may look at the definition for $\Re_2$, and show that a center is a solution of

$$\begin{pmatrix} x_1 & \cdots & x_n 1 \end{pmatrix} \begin{pmatrix} h_{11} & \cdots & h_{1n} \\ h_{1n} & \cdots & h_{nn} \\ p_1 & \cdots & p_n \end{pmatrix} = \begin{pmatrix} 0 & \cdots & 0 \end{pmatrix}$$

Here, the matrix is obtained from A by eliminating the last column.

Indeed, the problem of the existence of centers is the problem of existence of solutions of n linear equations in n unknowns. We shall only mention that if the determinant of Q is different from zero, then there is a unique center.

There are many interesting problems concerning quadrics. We shall state a few as exercises.

Exercise 5

(1) Change $3x^2 - 8xy + 7y^2$ to a canonical form, and obtain the matrix of change of basis.

(2) Given an $n \times n$ matrix A with real entries, show that the quadratic form $(A\xi, \xi)$ can be expressed in terms of a symmetric matrix.

(3) Give a condition for a straight line tangent to a quadric.

(4) Using Exercise 3, obtain tangent plane of a quadric at a given point on it.

(5) Define and obtain diametral plane of a quadric.

(6) Define and obtain principal planes of a quadric.

(7) Define and obtain vertices of a quadric.

(8) Define *pole* and *polar* with respect to a quadric, and obtain the polar of a point with respect to the quadric.

(9) Give several numerical examples in $\Re_2$.

(10) Do exercise (9) for $\Re_3$.

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6 Some Problems and Generalizations

In this chapter, we shall study a few ideas related to $\Re_2$, and then we try to generalize them. The purpose is acquiring the sense of generalizations.

6.1 Inversions

Let 0 be a fixed point, and $k > 0$ be a real number. An inversion with center 0 and power k is a function on the Euclidean plane onto itself, such that $f(A) = B$ means that 0, A, and B are co-linear and $(0A)(0B) = k$. We shall express this in terms of vectors. Let 0 end at the zero vector $\vec{0}$, A at ξ, and B at η. Then ξ and η are non-zero vectors, and $a > 0$, such that

$$\eta = a\xi$$

and

$$||\xi||\ ||\eta|| = k$$

This implies that

$$||\xi||\ |a|\ ||\xi|| = k$$

Thus

$$|a| = \frac{k}{||\xi||^2}$$

Thus the equation of this inversion will be

$$f(\xi) = \eta = \frac{k\xi}{||\xi||^2}, \qquad (1)$$
$$\xi \neq 0$$

In terms of coordinates, we may let

$$\xi = (x, y)$$

and

$$\eta = (x', y')$$

Thus (1) can be written as

$$\begin{cases} x' = \frac{kx}{x^2+y^2} \\ y' = \frac{ky}{x^2+y^2}, \\ x^2 + y^2 \neq 0 \end{cases} \qquad (2)$$

Indeed, (1) is independent of the dimension of the space. Thus, it can be defined as the equation of an inversion in $\Re_n$.

One may show that the inverse function of (1) exists, and also show that the circle $x^2 + y^2 = k$ is pointwise invariant under (2). We shall leave it as an exercise.

6.2 Quadric inversions

We shall generalize (2) to

$$\begin{cases} x' = \frac{kx}{ax^2+2bxy+cy^2} \\ y' = \frac{ky}{ax^2+2bxy+cy^2}, \\ ax^2+2bxy+cy^2 \neq 0 \end{cases} \tag{3}$$

Here we can easily see that the conic

$$ax^2 + 2bxy + cy^2 = k$$

is invariant under (3).

Now, we shall express (3) in terms of vectors and matrices. Note that

$$ax^2 + 2bxy + cy^2 \tag{4}$$

can be written as

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Let A be a linear transformation on $\Re_2$ whose matrix with respect to an orthonormal basis in $\Re_2$ is $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$, and $\xi = (x, y)$. Then the quadratic form can be written as $(A\xi, \xi)$. Thus (3) can be written as

$$\eta = \frac{k\xi}{(A\xi, \xi)},$$
$$(A\xi, \xi) \neq 0$$

where $\eta = (x', y')$.

Now we are in the position of defining a quadric inversion on E_n as a function on E_n defined by

$$f(\xi) = \frac{k\xi}{(A\xi, \xi)},$$
$$(A\xi, \xi) \neq 0$$

Here, A does not have to be Hermitian, and k may be a complex number.

6.3 Properties of quadric inversions

The quadric inversion with the equation

$$\eta = f(\xi) = \frac{p\xi}{A\xi, \xi)}, \tag{5}$$
$$(A\xi, \xi) \neq 0$$

transforms a hyperplane (co-set) to a quadric of the form

$$a(A\xi, \xi) + b(\xi, \delta) = 0 \tag{6}$$

which passes through the origin.

The proof is left as an exercise.

6.4 An inversion with center at α

In 6.1, if we move the center of inversion to the end point of the fixed vector α, the equation of the inversion will become

$$\eta = f(\xi) = \frac{k(\xi - \alpha)}{||\xi - \alpha||^2} + \alpha,$$
$$\xi - \alpha \neq \vec{0}$$

Indeed, when $\alpha = \vec{0}$, we obtain (1) of 6.1. We leave the proof as an exercise.

6.5 Theorem

An *inversion* is a conformal transformation; i.e., for non-zero $\xi, \eta \in E_n$, the value of $\frac{(\xi,\eta)}{||\xi|| \, ||\eta||}$ is invariant under an inversion.

6.6 Problem

Consider two circles (C_1) and (C_2) (Fig.9). Let the radii of (C_1) and (C_2) be r_1 and r_2 respectively.

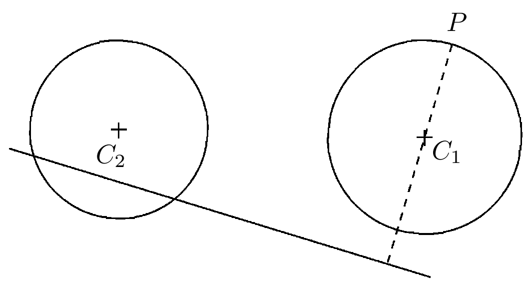

Fig. 9

Let P be a point on (C_1). Choose an inversion of center P and power equal to $PC_2^2 - r_2^2$. Show that (C_2) changes to a straight line. Show that the distance from C_1 to this straight line is independent of the choice of P.

The solution is left as an exercise.

6.7 Problem

Consider a circle of radius r, and a point 0 inside the circle (Fig. 10). Through 0, we draw two perpendicular lines.

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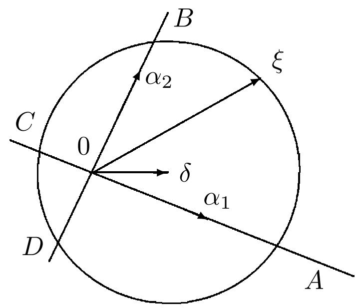

Fig. 10

They intersect the circle in A, B, C, and D. Show that

$$(0A)^2 + (0B)^2 + (0C)^2 + (0D)^2$$

is independent of the position of D and the orientation of the lines.

Solution

Let δ be a fixed vector ending at the center of the circle. The equation of the circle is

$$||\xi - \delta|| = r$$

Let $\{\alpha_1, \alpha_2\}$ be an orthonormal set such that α_1 is on the line $0A$, and α_2 is on the line $0B$. Thus the equation of the line $0A$ is

$$\xi = x\alpha_1$$

and the equation of the line $0B$ is

$$\xi = y\alpha_2$$

In order to obtain the point A and C, we must solve

$$\begin{cases} ||\xi - \delta|| = r \\ \xi = x\alpha_1 \end{cases}$$

Eliminating ξ, we obtain

$$x^2||\alpha_1||^2 - 2x(\delta, \alpha_1) + ||\delta||^2 - r^2 = 0 \tag{7}$$

Let x_1 correspond to A, and x_2 to C. Then

$$(0A) = x_1^2$$

and

$$(0C) = x_2^2$$

One observes that

$$x_1 + x_2 = (x_1 + x_2) - 2x_1x_2$$

where x_1 and x_2 are roots of (7). Thus

$$(0A)^2 + (0C)^2 = [2(\delta, \alpha_1)]^2 - 2||\delta||^2 + 2r^2$$

Similarly, for B and D, we must solve

$$\begin{cases} ||\xi - \delta|| = r \\ \xi = y\alpha_2 \end{cases}$$

This implies

$$(0B)^2 + (0D)^2 = [2(\delta, \alpha_2)]^2 - 2||\delta||^2 + 2r^2$$

Therefore

$$(0A)^2 + (0C)^2 + (0B)^2 + (0D)^2 = 4[(\delta, \alpha_1)^2 + (\delta, \alpha_2)^2] - 4||\delta||^2 + 4r^2$$

Note that (δ, α_1) and (δ, α_2) are components of δ with respect to the orthonormal set $\{\alpha_1, \alpha_2\}$. Thus

$$(\delta, \alpha_1) + (\delta, \alpha_2) = ||\delta||^2$$

This implies that

$$(0A)^2 + (0C)^2 + (0B)^2 + (0D)^2 = 4r^2$$

6.8 A generalization of 6.7

Consider the sphere $||\xi - \delta|| = r$ in E_n. Let $\{\alpha_1, \cdots, \alpha_n\}$ be orthonormal. Then

$$\begin{cases} ||\xi - \delta|| = r \\ \xi = x_i\alpha_i, \\ i = 1, \cdots, n \end{cases} \tag{8}$$

produces $2n$ vectors $n_1, \cdots, n_{2n}$, such that

$$\sum_{i=1}^{2n} ||\eta_i||^2$$

is independent of the orientation of $\{\alpha_1, \cdots, \alpha_n\}$.

To prove this, we follow the ideas of 6.7, and obtain

$$\sum_{i=1}^{n} (x_{i1}^2 + x_{i2}^n) = 2nr^2 + (4 - 2n)||\delta||^2$$

where solutions x_{i1}, x_{i2} correspond to two ηs on α_i. Note that if we put $n = 2$, we obtain the results of 6.6.

Exercise 6

(1) Study inversion for k being a complex number.

(2) In 6.2 (3), the denominator is a bilinear form. Generalize 6.2 (3) for a multilinear form.

(3) Study quadric inversion when k is a complex number and A is any linear transformation on E_n.

(4) Let A be a positive transformation on E. Consider the quadric inversion corresponding to A. Show that

$$\frac{(\xi,\eta)}{\sqrt{(A\xi,\xi)}\sqrt{(A\eta,\eta)}}$$

is invariant under this inversion.

(5) Generalize exercise (4) for any A and k.

(6) Consider the quadric inversion of 6.3 (5). Consider a vector η on which 6.3 (5) becomes inversion. Show that η is a proper vector of A.

(7) Generalize 6.7 to E_n.

(8) Generalize 6.7 using the space of linear transformations under the Frobenius inner product.

(9) Generalize exercise (8) using the space of linear transformation with $*$-products.

(10) Let A be a linear transformation on E_n such that

$$\frac{(A\xi,A\xi)}{||A\xi||\ ||A\xi||}=\frac{(\xi,\eta)}{||\xi||\ ||\eta||},$$
$$||A\xi||\neq 0,$$
$$||A\eta||\neq 0$$

for non-zero ξ and η in E_n. Then A is called a *conformal linear transformation*. Study A restricted to $\Re_2$. In general, show that A is conformal if and only if $A=cU$, where c is a complex number and U is a unitary transformation.

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